

# ABHISHEK BARAD

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## SUMMARY

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A M.S. scholar in Aerospace Engineering, expected to complete in June, 2026, with expertise in autonomous systems, specializing in Optimal Control theory, trajectory optimization, and collision avoidance. Proficiency in designing, developing, and validating autonomous guidance and control strategies for mobile and aerial robots. My background experiences in Electrical Engineering, and a strong background in control theory with programming (C++, Python) make me well-suited for robotics innovation projects.

## EDUCATION

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| <b>Indian Institute of Technology Madras</b> , Chennai, India                    | Jul 2023 - Present  |
| M.S. in Aerospace Engineering (Guidance, Navigation, and Control); CGPA: 8.18/10 |                     |
| <b>Veer Surendra Sai University of Technology</b> , Burla, India                 | Aug 2018 - May 2022 |
| B.Tech. in Electrical Engineering; CGPA: 8.25/10                                 |                     |

## EXPERIENCE

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| <b>Research Fellow</b>                                                                                                                                                                                                                                                                                                           | Jul 2023 - Present |
| Indian Institute of Technology Madras.                                                                                                                                                                                                                                                                                           |                    |
| <ul style="list-style-type: none"><li>– Assisted in analyzing and summarizing research findings for publication and presentations.</li><li>– Contributed to quizzes, preparing solutions, and grading assignments.</li><li>– Conducted comprehensive literature reviews and developed detailed research methodologies.</li></ul> |                    |

## AREAS OF INTEREST

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Linear Dynamical Systems      Nonlinear Dynamics and Control      Estimation Theory  
Optimization and Optimal Control

## RESEARCH AND ACADEMIC PROJECTS

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| <b>Final-Time Guidance for Terrain-Avoided Precision Soft Landing of Spacecraft</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Jan 2024 - Dec 2025 |
| (M.S. / Guide: Dr. Satadal Ghosh, Assistant Professor, IIT Madras)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                     |
| <ul style="list-style-type: none"><li>– <b>Built</b> an autonomous landing algorithm that <b>reduces fuel use</b> compared to fixed-time approaches.</li><li>– <b>Avoided hazards</b> using virtual barriers; maintained <b>125 m</b> median terrain clearance in simulations.</li><li>– <b>Respected</b> real-world engine limits via thrust bounds; ensured <b>robust, feasible</b> operation.</li><li>– <b>Optimized</b> landing duration per scenario ( <b>35–95 s</b> ) with a fast search routine.</li><li>– <b>Stress-tested</b> across <b>500</b> simulations under uncertainty; achieved consistent <b>precise, soft touchdowns</b>.</li></ul> |                     |

## SKILLS

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|-------------------------------|-------------|
| <b>Programming Languages:</b> | C++, Python |
| <b>Software/Frameworks:</b>   | Simulink    |

## CURRENTLY LEARNING

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|-----------------------------|-----------|
| <b>Tools:</b>               | PINNs, RL |
| <b>Software/Frameworks:</b> | ROS       |

## CONFERENCE PUBLICATIONS

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- **A. Barad** and S. Ghosh, “*A Fuel-Efficient Free-Final-Time Guidance for Terrain-Avoided Precision Soft Landing of Spacecraft*,” in 2025 IEEE 64th Conference on Decision and Control (CDC), pp. 948–953, 2025.

## HOBBIES AND OTHER INTERESTS

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Listening to music    Playing tennis    Watching movies